

## OPTIMIZATION · RECOMMENDATIONS · ALTERNATIVES

# Repair, or Rebuild

*The two delivered 2026–2027 call schedules, the reasoning behind them, and the options weighed along the way.*

**22**RULE & PTO CONFLICTS  
CLEARED**2**COMPLETE VERSIONS  
DELIVERED**0**CONFLICTS, PTO BREACHES  
& BACK-TO-BACKS LEFT**75**WEIGHTED LOAD NOW  
SHARED BY 10 OF 11  
SENIORS

**In brief.** One diagnosis of the draft schedule produced two finished answers — a **Minimal Repair** that touches only the rule violations, and a **Full Rebuild** optimized for fairness. This companion to the *Flagged-Dates Report* explains what each version is, exactly how they differ, the recommendations that hold no matter which you adopt, and the alternatives considered and set aside.

Companion to the Flagged-Dates Report · Both versions pass identical hard-rule QA

## S A Two delivered versions

**T**wo deliverables follow from one diagnosis, and they answer different questions. One asks “what is the smallest change that makes the draft legal?” The other asks “what is the fairest legal schedule we can build?” Both are complete, and both pass the same machine audit.

## VERSION 1

**Minimal Repair**

- Fixes **only** the rule violations — nothing else moves.
- Each fix is a 1-for-1 swap on the **same day of the week**.
- Every resident keeps their exact totals and day-mix.
- **Best for** adopting now with minimal review.

## VERSION 2

**Full Rebuild**

- Re-derives the **entire year** via CP-SAT optimization.
- Balanced so the on-call burden is as level as the rules allow.
- Totals re-balanced to near-equal within each class.
- **Best for** a fresh, provably-balanced calendar.

|                  | VERSION 1 — MINIMAL REPAIR                        | VERSION 2 — FULL REBUILD                      |
|------------------|---------------------------------------------------|-----------------------------------------------|
| GOAL             | Fix only the rule violations, change nothing else | Re-derive the whole year for maximum fairness |
| METHOD           | 1-for-1 same-day-of-week swaps                    | CP-SAT constraint optimization                |
| DISRUPTION       | 20 swaps; everything else identical to your draft | Most assignments re-derived                   |
| TOTALS & DAY-MIX | <b>Identical</b> to preliminary                   | Re-balanced near-equal within each class      |

0

ROTATION CONFLICTS (BOTH VERSIONS)

0

PTO CONFLICTS (BOTH VERSIONS)

0

DOUBLE-BOOKINGS (BOTH VERSIONS)

## § B Version 1 — Minimal Repair, in full

Each fix is a 1-for-1 swap with an eligible partner on the **same day of the week**, so both residents keep their exact call total and day-of-week distribution — only the dates move.

| #  | FIXED CONFLICT  | REASON                      | SWAP MADE                      |
|----|-----------------|-----------------------------|--------------------------------|
| 1  | 7/17 Bower      | on NICU                     | Patel ↔ Bower (Bower → 7/24)   |
| 2  | 10/30 Mautino   | on Away                     | Lux ↔ Mautino (→ 12/4)         |
| 3  | 11/13 Mautino   | on Away                     | Lux ↔ Mautino (→ 12/18)        |
| 4  | 4/14 Sublette   | on Peds ER                  | Bower ↔ Sublette (→ 4/7)       |
| 5  | 5/17 Bottke     | on Peds ER                  | Beutel ↔ Bottke (→ 5/24)       |
| 6  | 9/10 Patel      | Nexplanon (do-not-schedule) | Sublette ↔ Patel (→ 9/3)       |
| 7  | 3/15 Schutt     | PTO                         | Oehm ↔ Schutt (→ 3/8)          |
| 8  | 5/18 Wohlgemuth | PTO                         | Strand ↔ Wohlgemuth (→ 5/25)   |
| 9  | 8/25 Bottke     | Cardiology-Tue              | Sublette ↔ Bottke (→ 10/27)    |
| 10 | 1/12 Beutel     | Cardiology-Tue              | Bower ↔ Beutel (→ 12/29)       |
| 11 | 3/9 Gaspar      | Cardiology-Tue              | Bottke ↔ Gaspar (→ 4/27)       |
| 12 | 9/6 Sublette    | GYN/CM-Sun                  | Patel ↔ Sublette (→ 11/1)      |
| 13 | 9/27 Patel      | GYN/CM-Sun                  | Gaspar ↔ Patel (→ 12/13)       |
| 14 | 2/28 Kendrick   | GYN/CM-Sun                  | Bower ↔ Kendrick (→ 2/21)      |
| 15 | 3/17 Kendrick   | GYN/CM-Wed                  | Patel ↔ Kendrick (→ 2/24)      |
| 16 | 5/30 Bower      | GYN/CM-Sun                  | (3-way) → Gaspar; Bower → 5/16 |

| #         | FIXED CONFLICT | REASON                 | SWAP MADE                   |
|-----------|----------------|------------------------|-----------------------------|
| <b>17</b> | 11/4 Beutel    | <b>Ultrasound-Wed</b>  | Sublette ↔ Beutel (→ 11/11) |
| <b>18</b> | 11/11 Sublette | <b>Ultrasound-Wed</b>  | (paired with #17)           |
| <b>19</b> | 4/18 Kendrick  | <b>Ultrasound-Sun</b>  | Bower ↔ Kendrick (→ 4/25)   |
| <b>20</b> | 6/6 Gaspar     | <b>Ultrasound-Sun</b>  | Patel ↔ Gaspar (→ 5/30)     |
| <b>21</b> | 2/22 Mudondo   | <b>Dermatology-Mon</b> | Lux ↔ Mudondo (→ 3/8)       |
| <b>22</b> | 3/8 Lux        | <b>Dermatology-Mon</b> | (paired with #21)           |

#### WHY THE REPAIR IS INVISIBLE ON THE TALLY TABS

Every swap trades two residents who are both on call on the *same weekday*. Trading one Friday for another leaves each resident's count of Mondays, Tuesdays ... Sundays untouched — a permutation that preserves every marginal distribution. So the workbook's year-to-date totals are numerically identical before and after: twenty conflicts vanish without moving a single tally, and no new soft conflict or back-to-back call is introduced.

## § C Version 2 — Full Rebuild

The rebuild re-derives the year from the rules up, then makes it as even as those rules allow.

### RULES ENFORCED AS HARD CONSTRAINTS

- ✓ No resident on call during a **no-call rotation** (Away, NICU, Peds Inpatient, Peds ED, Leave).
- ✓ **Rotation day-restrictions:** GYN/CM no Sun/Mon/Wed · Ultrasound no Sun/Wed · Cardiology no Tue · Dermatology no Mon · FM-Ambulatory (2nd half) no Sun/Mon/Wed · R2-Pediatrics no Thu.
- ✓ **No CBP or R3-Orthopedics call at all** — even the “limited/soft” uses are eliminated.
- ✓ **PTO** windows and the **9/10 Nexplanon** date fully respected.
- ✓ **Night-float** weeknights fixed by the rotation grid; **no back-to-back** call (post-call day) for anyone.
- ✓ **R3 take no Sunday senior call** (single Lux-style exception permitted).
- ✓ **R2 carry 7–8 Sundays and 14–16 weekend (Fri+Sat+Sun) nights** each.
- ✓ **Holidays:** ≤ one major or two minor per senior; the four “mix-up” holidays on four different interns; interns cover Thanksgiving, Christmas & New Year's.
- ✓ **Coverage:** one intern + one senior every night, including the 12/20–1/2 winter block.

**THE FAIRNESS OBJECTIVE** Within each class the solver minimizes the inequality of every weekday's load and of the total weighted-inconvenience load, where each call is weighted by how unwelcome its day is:

weight Wed, Thu = **1** · Mon, Tue = **2** · Fri, Sun = **3** · Sat = **4**

The schedule is then made as even as the rules allow — lower spread is fairer.

### Balance, before vs after

“Δ” is the gap between the most- and least-burdened resident in the class.

| METRIC                                 | PRELIMINARY | REBUILD (V2)      |
|----------------------------------------|-------------|-------------------|
| Hard rotation conflicts                | 19          | <b>0</b>          |
| PTO / Nexplanon conflicts              | 3           | <b>0</b>          |
| CBP / R3-Ortho (“soft”) call           | 11          | <b>0</b>          |
| Back-to-back call (no post-call)       | 4           | <b>0</b>          |
| Interns covering Christmas / NYE / NYD | none        | <b>all 3</b>      |
| “Mix-up” holidays on distinct interns  | no          | <b>4 distinct</b> |
| R2 weighted-load spread (excl. Patel)  | 74–80 (Δ6)  | <b>75–75 (Δ0)</b> |

| METRIC                              | PRELIMINARY | REBUILD (V2) |
|-------------------------------------|-------------|--------------|
| R2 Saturdays (excl. Patel)          | 4–7 (Δ3)    | 4–5 (Δ1)     |
| R3 weighted-load spread             | 71–80 (Δ9)  | 75–75 (Δ0)   |
| R3 Saturdays                        | 4–6 (Δ2)    | 5–5 (Δ0)     |
| Intern totals                       | 55–58 (Δ3)  | 59–59 (Δ0)   |
| R3 Sunday calls (rule)              | Lux 1       | 0            |
| Patel (preserved outlier)           | 35          | 36           |
| Senior holiday-weight cap respected | no          | yes          |

In the rebuild the six non-Patel R2s share a near-identical mix (32–33 calls, **weighted load 75 each**) and the four R3s are **identical** (34 calls, weighted load 75 each) — so R2 and R3 now carry the *same* weighted load, with per-call inconvenience averages of 2.27–2.34 (R2) and 2.21 (R3). Patel is the single intentional outlier (36 calls, his 12 preserved backup days). No R3 works a Sunday at all — the rule's ideal — and the heaviest-loaded R3 (Mautino, formerly 9 Fridays + 6 Saturdays) is now level with peers.

## § D Three deliberate structural decisions

Three choices in the rebuild are intentional rather than incidental — worth stating plainly.

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Winter coverage</b> | Interns are added to the <b>12/20–1/2</b> holiday block so Christmas and New Year's have intern call — per the rule that interns schedule Thanksgiving, Christmas and New Year's. The preliminary had seniors covering solo.                                                                                                                                                                                                          |
| <b>Patel preserved</b> | Patel keeps his <b>12 early-year “second-senior” backup days</b> and elevated total (35–36 vs the class's 33), reflecting his off-cycle volume. Ten original dates are kept exactly; the two that were themselves rule violations — 9/10 (Nexplanon) and 9/27 (GYN/CM Sunday) — are relocated to other Patel-eligible July–September weekdays. He is the single intentional outlier; his backup-day partner rotates across the class. |
| <b>8/20 restored</b>   | The preliminary's missing intern on <b>8/20</b> (Hanna Hard's night-float block) is filled.                                                                                                                                                                                                                                                                                                                                           |

## § E Recommendations

Independent of which version you adopt, six actions carry the most value — roughly in priority order.

- 1 **Fix the five no-call-rotation errors first** — Bower 7/17, Mautino 10/30 & 11/13, Sublette 4/14, Bottke 5/17. Highest-risk: a resident physically unavailable for call.
- 2 **Add intern coverage to the winter holidays** (Christmas, New Year's) — currently senior-only.
- 3 **Even out holidays.** Stanek (3) and Beutel (3 minor) are over; Hard, Strand, Bower, Sublette are under. Put the four “mix-up” holidays on four different interns.
- 4 **Relieve Mautino** — in the preliminary he carries the heaviest senior load (9 Fridays + 6 Saturdays).
- 5 **Respect post-call** on the four back-to-back pairs, especially the two holiday-weekend ones.
- 6 **Add a block-boundary check.** Several conflicts fall on the first/last day of a 2-week rotation block (e.g., 9/27, 2/28) — worth a standing check whenever a rotation changes mid-weekend.

## § F Alternatives considered

### Pure fairness REJECTED

Letting R3s share Sundays makes every senior's day-mix identical — the tightest possible balance — but it breaks the R3-no-Sunday rule, so it was not used.

### Preserve the R2-Sunday / R3-Friday split ADOPTED

Slightly less uniform across classes but faithful to program structure. This is the Version 2 you have.

### Minimal vs full

Version 1 if you want the smallest change that is rule-clean; Version 2 if you want the fairest calendar. Both are delivered, so the choice stays open.

### WORKBOOK TRACKING TABLES — BOTH EXCEL VERSIONS

Both workbooks keep the program's original **live formulas** — the monthly per-resident COUNTIF totals, the “Day of Weeks Tracking” tables, and the YTD cross-month SUM rollups. None were replaced with static numbers. Because the schedule changed, every formula's cached value was recomputed and written back — so the tables read correctly on open — and then **verified cell-by-cell against the solved schedule**: all 17 residents' totals and all five day-of-week groups match, on every month tab and the YTD tab.

## :: At a glance

|               |                                                               |                                            |                                                            |
|---------------|---------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|
| <b>19 + 3</b> | Rotation + PTO conflicts found in the draft<br>— all cleared  | <b>20</b>                                  | Same-day-of-week swaps in the Minimal Repair               |
| <b>75</b>     | Weighted load carried by ten of eleven seniors in the rebuild | <b><math>\Delta 9 \rightarrow 0</math></b> | R3 weighted-load spread, preliminary $\rightarrow$ rebuild |
| <b>12</b>     | Patel backup days preserved (10 kept, 2 relocated)            | <b>0</b>                                   | Conflicts, PTO breaches & back-to-backs in both versions   |

**About these versions.** Both are delivered as live workbooks in the original 16-tab format and as standalone, interactive calendars. **Version 1 (Minimal Repair)** is the smallest rule-clean change to your draft; **Version 2 (Full Rebuild)** is the fairness-optimized calendar (Google OR-Tools CP-SAT). See the companion *Methodology* document for the mathematics, and the *Flagged-Dates Report* for the full list of issues in the draft. · UnityPoint Health Family Medicine Residency · Academic Year 2026–2027.